

Dongchu XIE

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EDUCATION

The Chinese University of Hong Kong, Shenzhen

Shenzhen, China

Bachelor of Engineering in Computer Science and Engineering

September 2023 – Present

Expected Graduation: May 2027

CGPA: 3.572/4.00

Correspondence Address: Shaw College, The Chinese University of Hong Kong, Shenzhen, 2001 Longxiang Boulevard, Longgang District, Shenzhen 518172, P. R. China

Selected Coursework: Calculus, Probability and Statistics, Linear Algebra, C/C++ Programming, Machine Intelligence and Applications, Machine Learning, Reinforcement Learning, Human–Computer Interaction, Computer Networks, Robotics, and Embedded Systems

PUBLICATIONS

1. Hanfu-Bench: A Multimodal Benchmark on Cross-Temporal Cultural Understanding and Transcreation

Li Zhou, Lutong Yu, **Dongchu Xie**, Shaohuan Cheng, Wenyan Li, and Haizhou Li. *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing (EMNLP 2025)*, pp. 24616–24638, 2025. doi:10.18653/v1/2025.emnlp-main.1251.

2. Emo-LiPO: Listwise Preference Optimization for Fine-Grained Emotion Intensity Control in LLM-based Text-to-Speech

Yihang Lin, Li Zhou, Congwei Cao, **Dongchu Xie**, Xiaoxue Gao, Chen Zhang, and Haizhou Li. Accepted at the *International Joint Conference on Artificial Intelligence (IJCAI 2026)*, 2026. doi:10.48550/arXiv.2606.13006.

3. Domain-Explained Residuals for Robust Offline-to-Online Reinforcement Learning under Dynamics Shift and Data Corruption

Dongchu Xie, Xingrui Yu, Zhenglin Wan, and Chubin Zhang. Submitted to the *41st AAAI Conference on Artificial Intelligence (AAAI 2027)*; under review.

4. HSSR-LM: Descriptor-Level Translation between Porous Crystal Representations and Natural Language

Dongchu Xie and Chao Xue. Submitted to the *41st AAAI Conference on Artificial Intelligence (AAAI 2027)*; under review.

RESEARCH AND PROFESSIONAL EXPERIENCE

Junior Research Assistant

September 2026 – Present

Department of Chemical and Biological Engineering, The Hong Kong University of Science and Technology
Hong Kong, China

- Contribute to AI-for-Science foundation-model research, focusing on continued pre-training design and implementation.
- Acquire and process multimodal scientific data—including molecules, proteins, materials, RNA, and DNA—and design modality-specific tokenization methods.

Part-time Research Intern

November 2025 – July 2026

*Centre for Frontier AI Research, Agency for Science, Technology and Research (A*STAR)* Singapore

- **Domain-Explained Residuals (AAAI 2027 submission):** Designed robust algorithms to address dynamics shifts and severe data corruption, improving stability from adversarial offline datasets through online fine-tuning and deployment.
- **HSSR-LM (AAAI 2027 submission):** Developed a bidirectional translation system between porous crystal structures (MOFs and zeolites) and natural language; proposed the Hierarchical Structure Sequence Representation and curated 8,000 multi-granularity structure–text pairs.

Research Trainee, Chemical Engineering

May 2025 – July 2025

TU Dortmund University

Dortmund, Germany

- Developed and evaluated a neural-network-based approximate model predictive controller for a constrained continuous stirred-tank reactor, delivering a functioning prototype and final technical presentation.

Part-time Research Assistant

April 2025 – August 2025

Urban Governance and Design Thrust, The Hong Kong University of Science and Technology (Guangzhou)

Guangzhou, China

- Synthesized literature on computational sociology and urban governance, cleaned and integrated multi-source datasets, and co-authored the resulting manuscript.

Part-time Research Assistant

January 2025 – January 2026

Human Language Technology Lab, The Chinese University of Hong Kong, Shenzhen

Shenzhen, China

- **Hanfu-Bench (EMNLP 2025):** Co-developed a multimodal cultural-understanding benchmark; curated 1,192 expert-annotated images spanning five dynasties and designed cultural visual-understanding tasks.
- **Emo-LiPO (IJCAI 2026):** Proposed listwise preference optimization for fine-grained emotion-intensity control in TTS, constructed the ESD-plus dataset, and outperformed DPO-based baselines in controllability.

LEADERSHIP AND SERVICE

President, Model United Nations Association, CUHK-Shenzhen September 2024 – September 2025

- Led a 24-member executive team and organized the university's annual Model UN conference for more than 150 participants from across the Asia-Pacific; oversaw strategy, budget, logistics, recruitment, and external partnerships.

President, TEDxCUHKSZ

September 2024 – September 2025

- Led a 16-member executive team and delivered an annual TEDx conference for over 200 attendees, overseeing speaker curation, venue design, volunteers, sponsorships, and institutional partnerships.

Dais Member, Greater Bay Area Model United Nations 2025

March 2025

- Authored background guides, moderated committee proceedings, evaluated draft resolutions, mentored over 100 delegates, and coordinated cross-institutional committee operations while representing CUHK-Shenzhen.

Keynote Speaker, Student Sub-forum of the 2024 Westlake Forum on Higher Education October 2024

- Invited by Westlake University to discuss university life, global competitiveness, and pathways for youth development with students from leading international universities, including Harvard and Yale.

Volunteer Educator, Shenzhen Foreign Languages School

September 2024 – December 2024

- Initiated and delivered a global-citizenship program integrating Model UN methods, international relations, cross-cultural communication, diplomatic negotiation, and mock conferences.

Volunteer, 2024 Volvo China Open Organizing Committee

March 2024 – May 2024

- Managed spectator flow and course protocols and supported crowd safety, route management, and on-site logistics for an international professional golf tournament.

HONORS AND AWARDS

Honorable Mention, 2026 Mathematical Contest in Modeling — COMAP May 2026

Undergraduate Research Award — The Chinese University of Hong Kong, Shenzhen April 2026

Dean's List — School of Science and Engineering, CUHK-Shenzhen May 2025

Shaw Spirit Award — Shaw College, CUHK-Shenzhen November 2024

Shandong Provincial Outstanding Student Cadre of Senior High School — Department of Education of Shandong Province April 2023

Shandong Provincial Outstanding High School Student — Department of Education of Shandong Province May 2021